

Early Project Report

Seismic facies clustering using Torque clustering

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Project Goal

The goal of this project is to develop, test, and analyze clustering methods for seismic facies segmentation, with the core implementation centered on Torque Clustering and its extensions [7]. The final objective is to obtain robust unsupervised facies maps that are geologically consistent and useful for interpretation workflows.

Seismic Data

In seismic acquisition and processing, each sample can be represented by multiple attributes (amplitude, phase/frequency, geometry, and texture/coherence-like descriptors). Using these multi-attribute vectors, classical clustering methods can already produce useful first-order facies separation.

Our project focuses on multi-attribute clustering where each point (or patch-level feature) is represented in feature space and segmented via graph-based and hierarchical logic.

Project Plan

1. Baseline implementation and validation

Implement and validate Torque Clustering and CNS-based modules on synthetic datasets with full stage-wise visualizations [7,8].

2. Hybrid optimization model

Implement an ADMM-based CNS+Torque model, where graph smoothness, prototype reconstruction, edge regularization, and consensus constraints are optimized jointly.

3. Seismic application phase

Transfer validated methods to seismic attribute data; compare resulting facies maps to baselines (K-means, fuzzy clustering, and spectral-style pipelines).

4. Evaluation

Evaluate stability, spatial continuity, geological plausibility, and sensitivity to hyperparameters.

Literature Review

Classical and unsupervised facies clustering

Seismic facies analysis has a long tradition of attribute-based unsupervised learning. K-means, hierarchical clustering, and fuzzy clustering are common baselines [1,2].

Methods used in this project

We use and extend:

- **Torque Clustering (TC):** hierarchical connection-based clustering with torque-like scoring and tGap-based cut selection [7].
- **Clustering by Nonparametric Smoothing (CNS):** nonparametric smoothing-based soft assignment and model selection [8].
- **New ADMM hybrid:** joint CNS+TC optimization in one objective:

$$\min_{P, Z, U, V} \alpha \text{Tr}(P^\top L P) + \beta \|X - ZU^\top\|_F^2 + \gamma \sum_e c_e \|V_e\|_2 + \lambda \|P - Z\|_F^2, \quad \text{s.t. } P, Z \in \Delta, V = BZ.$$

Expected Outcomes

- Reproducible software pipeline for TC, CNS, and ADMM-based hybrid clustering.
- Stage-wise visual diagnostics of optimization and clustering behavior.
- Comparative study on synthetic and seismic-attribute datasets.
- Practical recommendations for method selection by geometry and data regime.

References

1. E. Y. Oliveira et al., “K-means clustering using principal component analysis to automate label organization in multi-attribute seismic facies analysis,” *Journal of Applied Geophysics*, 2022.
2. X. Wang et al., “Unsupervised seismic facies analysis with spatial constraints using regularized fuzzy c-means,” *Journal of Geophysics and Engineering*, 2017.
3. K. Li et al., “CONSS: Contrastive Learning Approach for Semi-Supervised Seismic Facies Classification,” arXiv:2210.04776, 2022.
4. G. C. Nwankwo et al., “Deep semi-supervised learning using generative adversarial networks for automated seismic facies classification of mass transport complex,” *Computers & Geosciences*, 2023.

5. H. Di and A. Abubakar, “Automating seismic-well tie via self-supervised learning,” *Geophysical Prospecting*, 2023.
6. J. Zhao et al., “Intelligent recognition of seismic facies based on contrastive semi-supervised learning,” *Geophysical Prospecting for Petroleum*, 2024.
7. J. Yang and C.-T. Lin, “Autonomous clustering by fast find of mass and distance peaks,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 47, no. 7, pp. 5336–5349, 2025, doi: 10.1109/TPAMI.2025.3535743.
8. D. P. Hofmeyr, “Clustering by Nonparametric Smoothing,” arXiv:2503.09134, 2025.